CurricuPath Pilot Evaluation Report

Project 26P73: AI-assisted Wits EIE learning pathway and module advisor

Evaluation protocol v2.5, pilot runs | 24 September 2026 | code version 5854b3df

Status: pilot, not a validated result. No independent academic reviewer has approved reference labels yet. Every score below is measured against developer-side answer keys, so it describes how the systems behave and differ. It is not evidence of independently validated accuracy. Figures published for other systems appear as context only.

Summary

- **What was run.** Five systems answered the same 60 synthetic EFA03/EFA04 eligibility cases once each, and the deterministic intent extractor was tested on 100 draft prompts.
- **CurricuPath against the handbook-RAG baseline.** Macro-F1 is 0.89 against 0.71 under the developer labels (difference +0.18, 95% interval +0.03 to +0.32; CurricuPath higher). Under the Senate Rules briefing labels it is 0.65 against 0.51 (inconclusive).
- **The answer key matters.** The two developer-side label sets agree on 78% of cases (kappa 0.67) and differ on 13, all on whether a corequisite that is only *planned* counts as satisfied. CurricuPath's macro-F1 moves from 0.89 to 0.65 between them.
- **CurricuPath makes the same decisions as the rules-only system on all 60 cases.** Any advantage over rules-only must come from language handling and clarification, which this case set does not test.
- **Intent understanding:** 85/100 correct (85%). The weak categories are career-path prompts and ambiguous module names.

1. Purpose and scope

The investigation asks whether hard eligibility gates and conversational clarification improve the correctness of Wits EIE curriculum advice compared with simpler strategies. The study is limited to eligibility decisions for the EFA03 (Electrical Engineering) and EFA04 (Information Engineering) programmes, each returned as one of three outcomes: *eligible*, *ineligible* or *manual review required*.

Under the frozen protocol (section 16), work without a named independent reviewer may continue only as a pilot and must not claim independently validated accuracy. This report is that pilot. It shows the evaluation pipeline working end to end on real system outputs and reports what it finds.

2. Method

2.1 Systems compared

System	What it does	Pilot configuration
CurricuPath	Intent extraction, module-name resolution, deterministic eligibility gates, and one scripted clarification where permitted.	Deterministic extraction; Gemini not used.
CurricuPath (no clarification)	The same system with clarification switched off.	Deterministic extraction; Gemini not used.
Rules only	The deterministic gates applied directly to the target module. No language layer.	No model.
Handbook RAG	Retrieves the 20 most similar handbook passages and asks one model to decide, with no rules supplied. The baseline representing existing handbook chatbots.	gemini-3.5-flash-lite, top-k 20.
Structured-rules LLM	As Handbook RAG, but the structured rules and source states are added to the prompt.	gemini-3.5-flash-lite, top-k 20.

2.2 Cases

60 synthetic profiles (no real student data), all drafted by the developers: 20 eligible, 20 ineligible and 20 manual review. Scenario types: academic discretion or unresolved record (20), corequisite unknown (10), prerequisite missing (10), prerequisite satisfied (20). Ten cases permit one scripted clarification.

2.3 Answer keys

Two developer-side keys were used. Neither is independent evidence.

- **Developer labels:** the expected outcome written when each case was drafted.
- **Senate Rules briefing labels:** a second reading of the same cases from a briefing document built from the 2026 Senate Rules. The document calls every label a draft and records no reviewer, so it is used only as a sensitivity check.

2.4 Measures

Three-class macro-F1 and balanced accuracy, with 95% intervals from a case-level bootstrap (5000 resamples, seed 20260918), and paired differences that share each resample. A difference is called directional only when its interval excludes zero; otherwise it is reported as inconclusive. A run with no valid decision counts as an error and stays in every denominator. Error types are reported separately: an unsafe false positive is a case that was not eligible but was answered eligible.

2.5 How the pilot departs from the frozen protocol

- One run per case instead of three, so repeatability is not measured.
- LLM arms used gemini-3.5-flash-lite, not the pinned gemini-3.6-flash, because of quota.
- CurricuPath arms ran without Gemini intent extraction, so they show the deterministic path only.
- An earlier Handbook-RAG attempt lost 27 of 60 runs to a per-minute quota limit; it was re-run paced and all 60 are valid.

3. Eligibility results

3.1 Agreement with the answer keys

System	Macro-F1, developer labels (95% CI)	Balanced accuracy	Macro-F1, briefing labels (95% CI)	Balanced accuracy	No valid decision
CurricuPath	0.89 (0.81 to 0.95)	0.83	0.65 (0.51 to 0.76)	0.70	10
Handbook RAG	0.71 (0.60 to 0.81)	0.72	0.51 (0.39 to 0.62)	0.54	0
Structured-rules LLM	0.80 (0.70 to 0.89)	0.80	0.66 (0.53 to 0.78)	0.77	0
Rules only	0.89 (0.81 to 0.95)	0.83	0.65 (0.51 to 0.76)	0.70	10
CurricuPath (no clarification)	0.89 (0.81 to 0.95)	0.83	0.65 (0.51 to 0.76)	0.70	10

Rules only and both CurricuPath variants return no valid decision on 10 cases. In those *corequisite unknown* cases the profile does not declare the corequisite, and the engine declines to decide rather than assume the module is not taken. The developers labelled them *ineligible*, so the protocol counts each as an error.

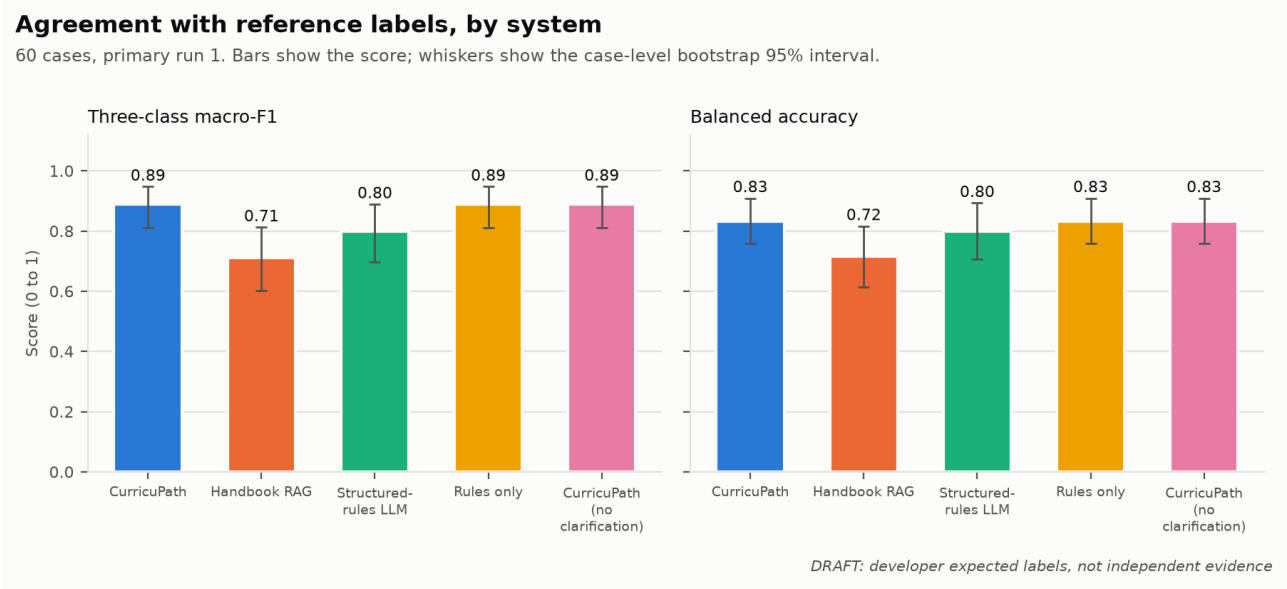


Figure 1. Agreement with the developer labels. Whiskers show the bootstrap 95% interval.

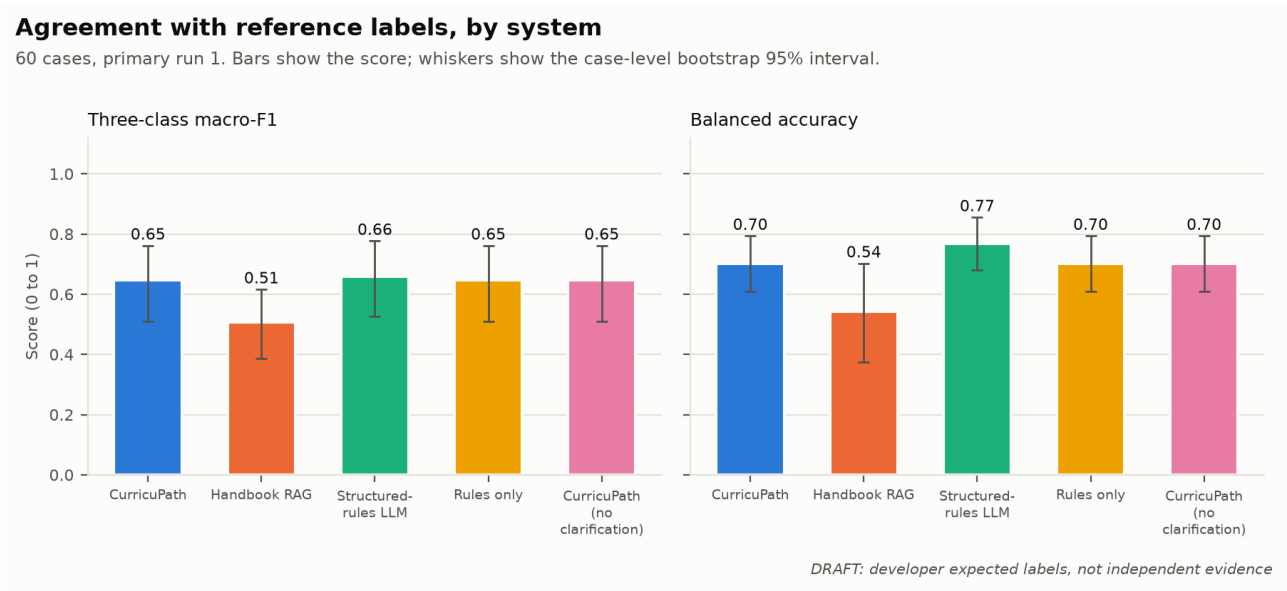


Figure 2. Agreement with the Senate Rules briefing labels.



Figure 3. Confusion matrices against the developer labels. Rows are the reference label.

3.2 Paired differences

Macro-F1 difference between CurricuPath and each other system on the same cases.

CurricuPath minus	Developer labels: difference (95% CI)	Reading	Briefing labels: difference (95% CI)	Reading
Handbook RAG	+0.18 (+0.03 to +0.32)	higher	+0.14 (-0.01 to +0.28)	inconclusive
Structured-rules LLM	+0.09 (-0.04 to +0.22)	inconclusive	-0.01 (-0.12 to +0.09)	inconclusive

CurricuPath minus	Developer labels: difference (95% CI)	Reading	Briefing labels: difference (95% CI)	Reading
Rules only	+0.00 (+0.00 to +0.00)	inconclusive	+0.00 (+0.00 to +0.00)	inconclusive
CurricuPath (no clarification)	+0.00 (+0.00 to +0.00)	inconclusive	+0.00 (+0.00 to +0.00)	inconclusive

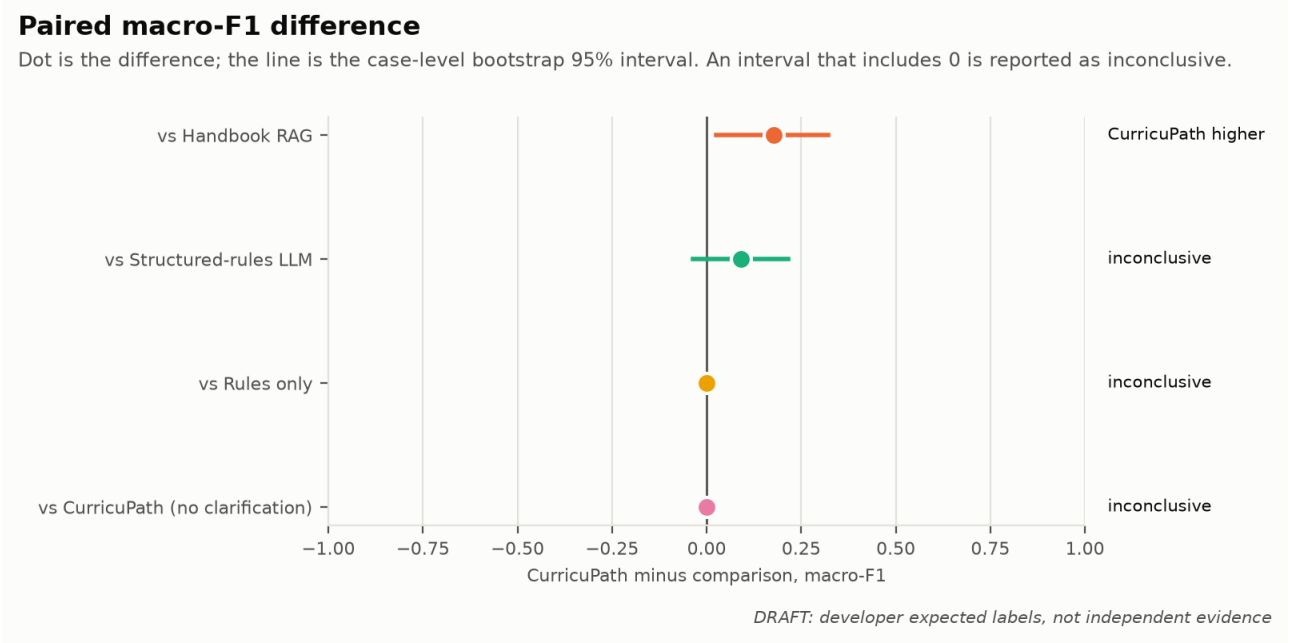


Figure 4. Paired macro-F1 differences against the developer labels.

3.3 Error types

Against the developer labels:

System	Unsafe false positive	Overly conservative	Correct escalation	Unsafe overconfidence	Unnecessary escalation
CurricuPath	0% (0/40)	0% (0/20)	100% (20/20)	0% (0/20)	0% (0/40)
Handbook RAG	28% (11/40)	30% (6/20)	100% (20/20)	0% (0/20)	0% (0/40)
Structured-rules LLM	12% (5/40)	30% (6/20)	100% (20/20)	0% (0/20)	2% (1/40)
Rules only	0% (0/40)	0% (0/20)	100% (20/20)	0% (0/20)	0% (0/40)
CurricuPath (no clarification)	0% (0/40)	0% (0/20)	100% (20/20)	0% (0/20)	0% (0/40)

Against the Senate Rules briefing labels:

System	Unsafe false positive	Overly conservative	Correct escalation	Unsafe overconfidence	Unnecessary escalation
CurricuPath	25% (13/53)	0% (0/7)	61% (20/33)	39% (13/33)	0% (0/27)
Handbook RAG	40% (21/53)	43% (3/7)	61% (20/33)	39% (13/33)	0% (0/27)
Structured-rules LLM	23% (12/53)	0% (0/7)	61% (20/33)	39% (13/33)	4% (1/27)
Rules only	25% (13/53)	0% (0/7)	61% (20/33)	39% (13/33)	0% (0/27)
CurricuPath (no clarification)	25% (13/53)	0% (0/7)	61% (20/33)	39% (13/33)	0% (0/27)

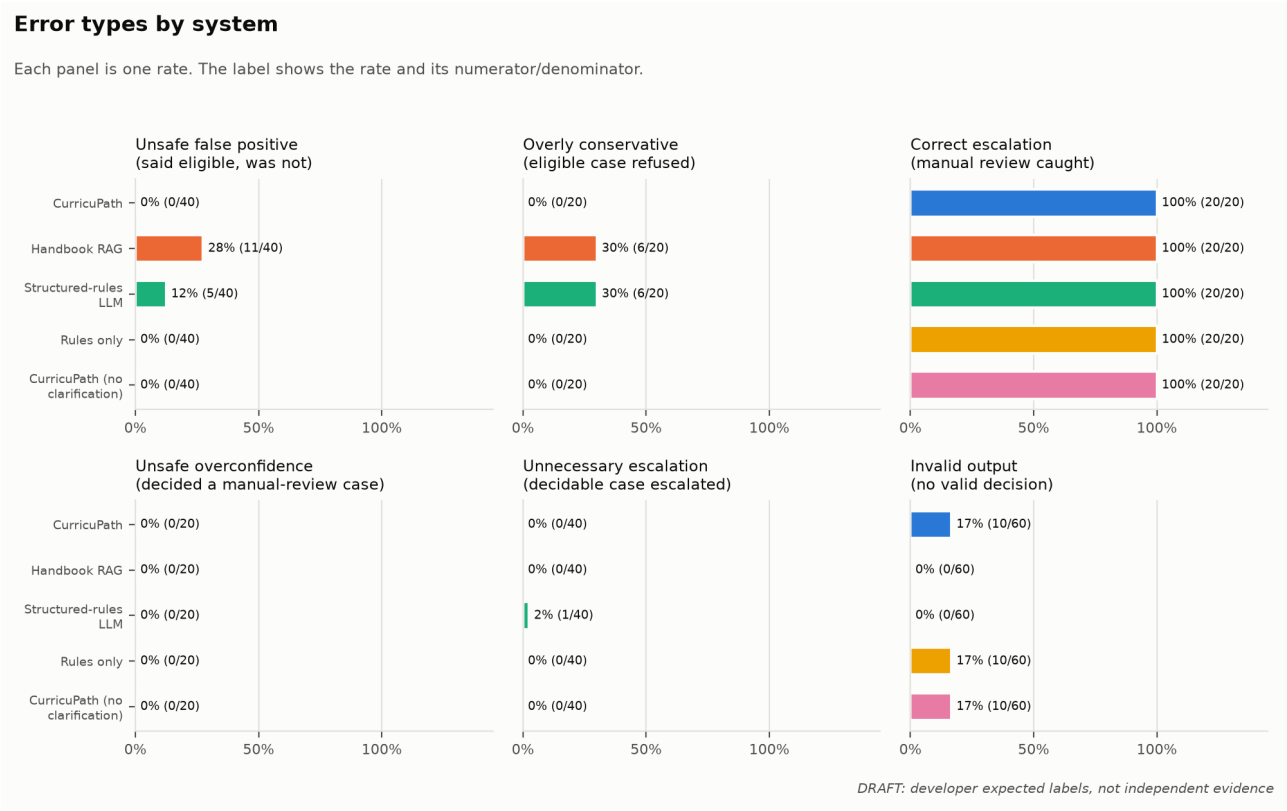


Figure 5. Error types against the developer labels.

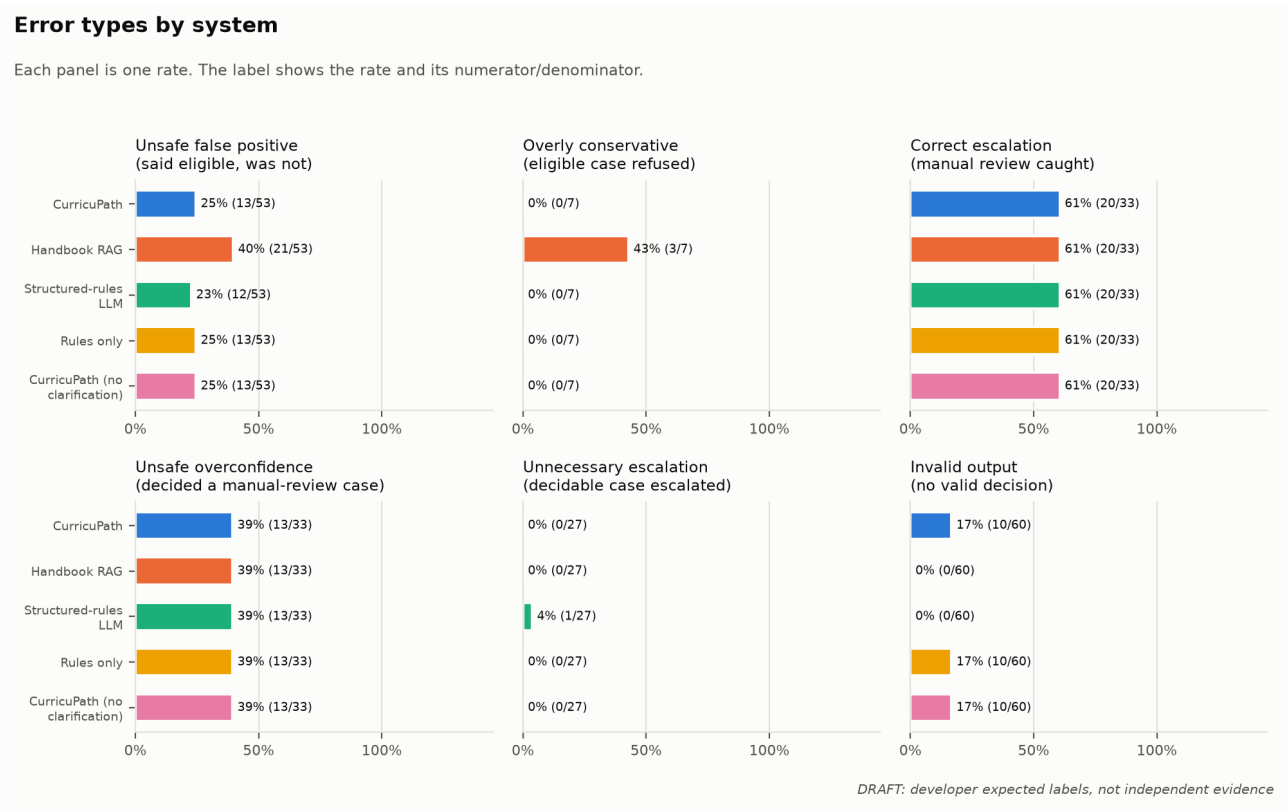


Figure 6. Error types against the Senate Rules briefing labels.

3.4 Sensitivity to the answer key

The two label sets agree on 78% of the 60 cases (Cohen's kappa 0.67). All 13 differences are the same: the developers marked the case *eligible* and the briefing marked it *manual review required*, because a corequisite appears only under the student's planned modules. Senate Rule 1.9 defines a corequisite as a course that must be taken with another, so whether a planned module counts is a question for an academic authority. The engine

currently accepts planned modules as satisfying a corequisite.

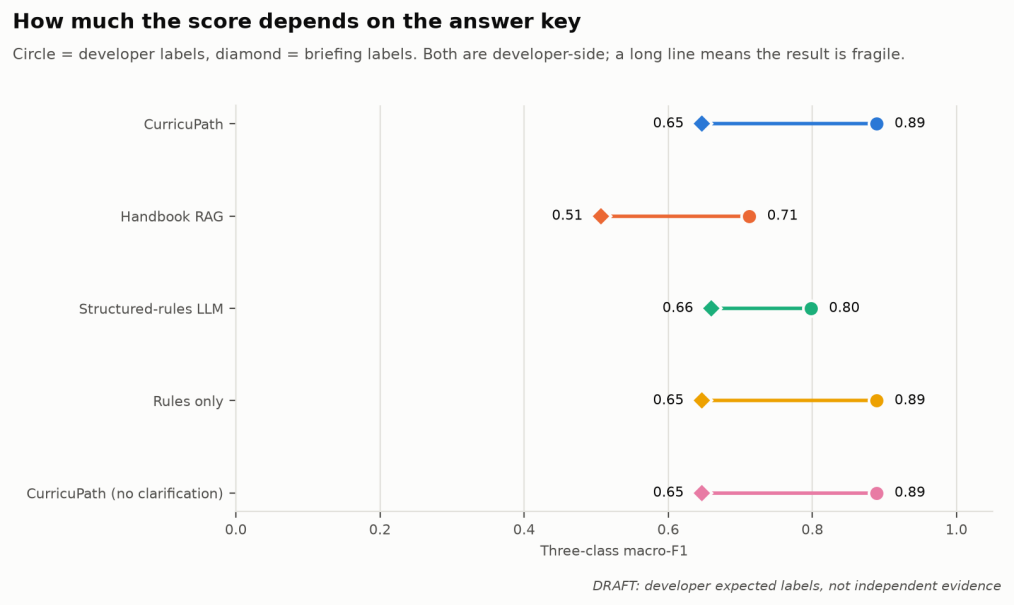


Figure 7. Macro-F1 under each answer key (circle: developer, diamond: briefing).

Cases where the label sets differ: ELIG001, ELIG002, ELIG003, ELIG005, ELIG006, ELIG007, ELIG010, ELIG013, ELIG014, ELIG015, ELIG018, ELIG019, ELIG020.

3.5 Response time

Times are per case, measured end to end through each system. The CurricuPath figures are for the deterministic path only; the Gemini-assisted path was not measured.

System	Median	95th percentile	Runs
CurricuPath	0.006 s	0.009 s	60
Handbook RAG	1.606 s	1.932 s	60
Structured-rules LLM	1.823 s	2.204 s	60
Rules only	under 1 ms	under 1 ms	60
CurricuPath (no clarification)	0.005 s	0.007 s	60

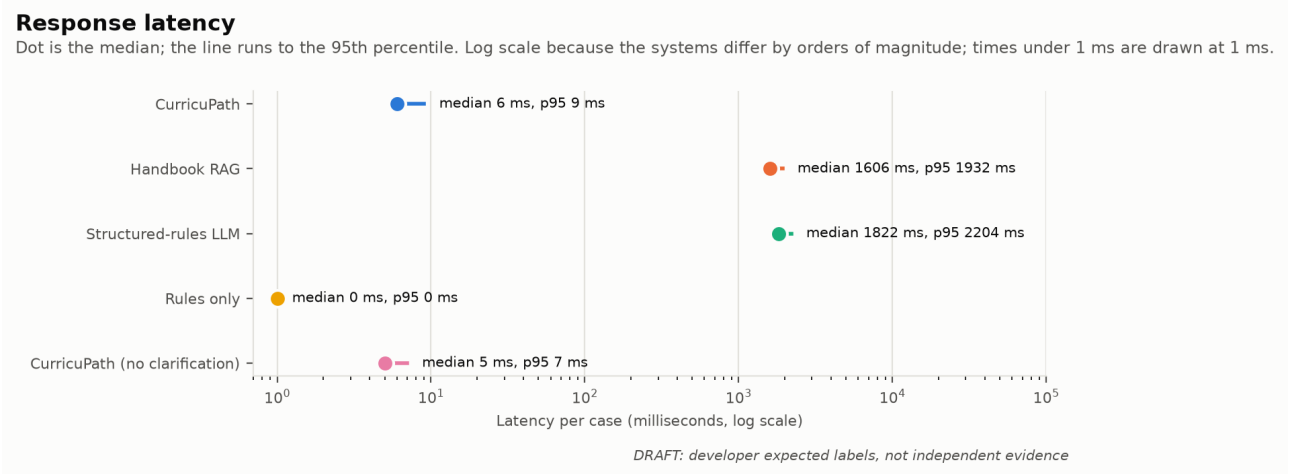


Figure 8. Response time by system (log scale).

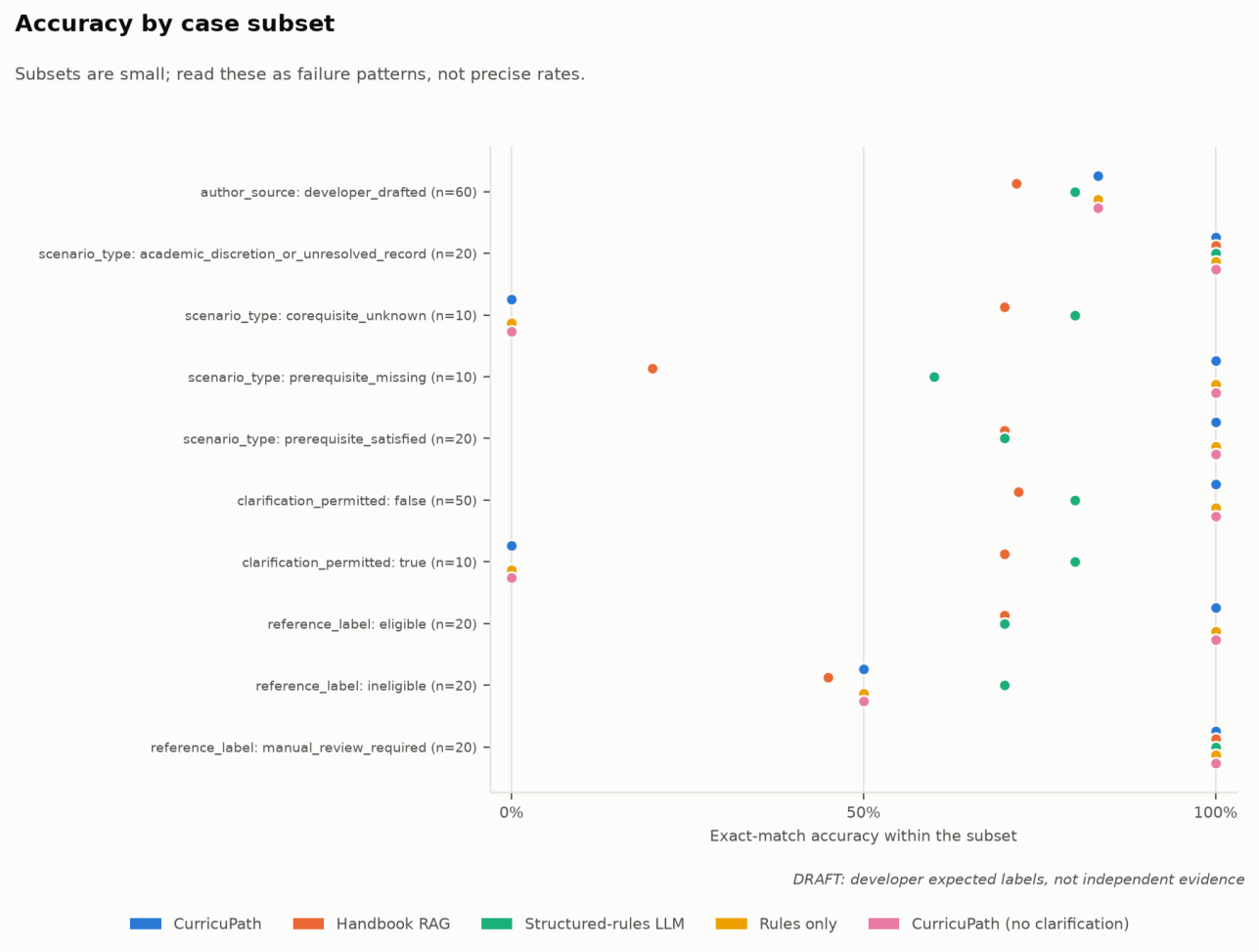


Figure 9. Exact-match accuracy within case subsets. Subsets hold 10 to 60 cases, so read these as failure patterns rather than precise rates.

4. Intent understanding

The deterministic extractor was run on 100 developer-drafted prompts that have not been independently reviewed. The Gemini-assisted extractor was not measured.

Prompt category	Correct	Rate	95% interval
ambiguous module name	0/10	0%	0.00 to 0.28
career path	6/10	60%	0.31 to 0.83
unknown or out of scope	4/5	80%	0.38 to 0.96
clarification follow up	5/5	100%	0.57 to 1.00
conversation control	5/5	100%	0.57 to 1.00
curriculum list	15/15	100%	0.80 to 1.00
eligibility	20/20	100%	0.84 to 1.00
module information	15/15	100%	0.80 to 1.00
public course search	15/15	100%	0.80 to 1.00
All prompts	**85/100**	**85%**	0.77 to 0.91

Where an entity was expected it was always extracted correctly: module code 35/35, programme code 18/18, year of study 12/12.

The ten *ambiguous module name* prompts (for example a topic such as electronics rather than a module code) are resolved by a later module-resolution layer, not by the extractor, so 0/10 understates the full system. The clearer weakness is career-path prompts, four of ten of which were treated as course searches.

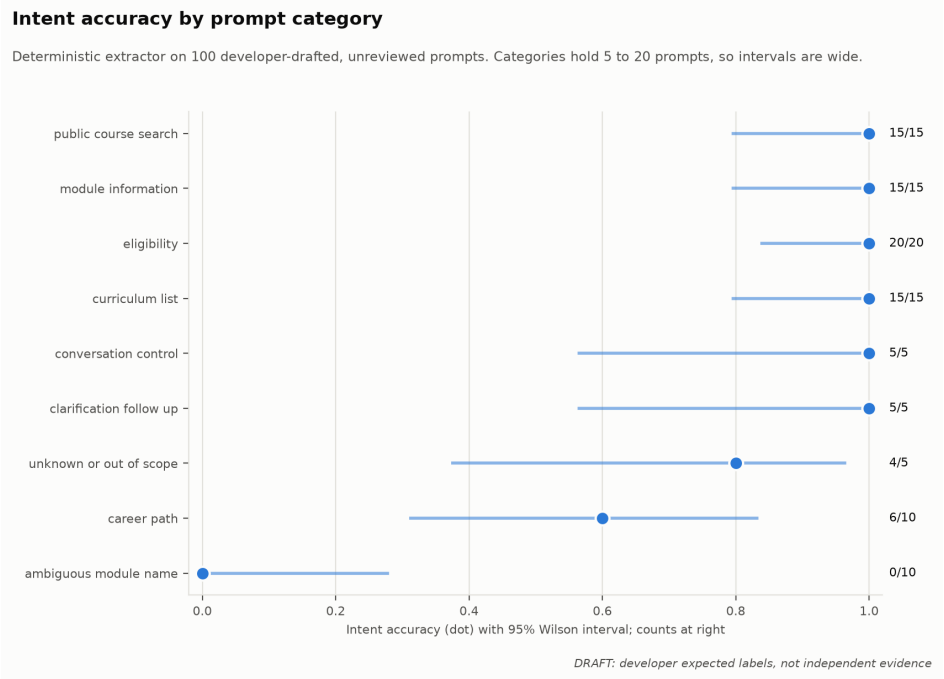


Figure 10. Intent accuracy by prompt category with 95% Wilson intervals.

5. Comparison with published benchmarks

Published systems used other institutions, datasets and definitions, so their figures cannot be reproduced here. Figure 11 places our measurements beside the matching published value. Filled marks are measured here; hollow marks are published. Read the published marks as context, not as parity.



Figure 11. Pilot measurements against published benchmarks.

Published figures with their sources

Benchmark	Metric	Published value	Data behind it
B6 NileAdvisor (Alaa et al. 2025)	average query time	2.07 (seconds)	single institution; Gemini to Cypher over Neo4j
B6 NileAdvisor (Alaa et al. 2025)	f1	0.958 (proportion)	300 processed queries listed (420 described); 3 tracks
B6 NileAdvisor (Alaa et al. 2025)	query success	0.93 (proportion)	curated query collection; single institution
B1 Islam and Hosen (2025)	prerequisite classification	0.99 (proportion)	synthetic: 101330 interactions, 10000 simulated students; reported as above 99%
B5 ARGObot (Tamascelli et al. 2025)	answer correctness	0.878 (proportion)	18 handbook questions

Benchmark	Metric	Published value	Data behind it
B5 ARGObot (Tamascelli et al. 2025)	faithfulness	0.927 (proportion)	18 handbook questions
B4 MyAdvisor (Kuhail et al. 2023)	student task success	0.92 (proportion)	17 students and 4 advisors
B4 MyAdvisor (Kuhail et al. 2023)	sus students	86.47 (score 0 100)	17 students
B3 Course recommender chatbot (Tan et al. 2022)	usability acceptance	8.2 (score 0 10)	35-student requirements survey
B2 Prerequisite-based recommender (Chanaa and El Faddouli 2024)	ndcg at 10	0.8648 (proportion)	public learning objects; not institutional rules

6. Findings

- **CurricuPath outperforms the handbook-RAG baseline under the developer labels** (+0.18 macro-F1, interval excludes zero), and points the same way under the briefing labels (+0.14) without reaching a directional result. The direction is stable; the strength depends on the answer key.
- **Against the structured-rules LLM the comparison is inconclusive under both keys** (+0.09 and -0.01). Giving a model the rules closes most of the gap on these cases, which supports the value of structured curation but does not by itself show that deterministic evaluation caused the gain.
- **CurricuPath and the rules-only system decide identically on all 60 cases.** These cases use clean, fixed wording, so they cannot show the value of language handling. That has to be shown by the ambiguity and clarification experiment.
- **The clarification step did not resolve any case in this run.** All 10 cases where clarification was permitted ended unresolved. The scripted reply is a general statement that every module has been listed, and the engine does not read that as declaring the missing corequisite absent. The frozen cases were not edited after seeing system output.
- **Whether planned corequisites count changes the unsafe-error picture.** Under the briefing reading, CurricuPath answers *eligible* on 13 of 53 cases that were not eligible (25%), against 0 under the developer labels. This is the question to put to the academic reviewer.
- **Handbook RAG shows the most unsafe errors** in both keys (28% and 40% of non-eligible cases answered eligible).

7. Limitations

- Both answer keys are developer-side. The cases and the rules they test were written by the same people, so agreement is partly circular.
- One run per case, so run-to-run repeatability is not measured. The model arms used a lighter model than the pinned one.
- CurricuPath ran without Gemini intent extraction. Its Gemini-assisted behaviour and response time are not measured here.
- 60 cases with about 20 per class give wide intervals. The study can show large failure patterns, not small improvements.
- The intent prompts are unreviewed developer drafts.
- Not yet evaluated: grounding of explanations, the clarification ablation on ambiguous cases, usability with students and advisors, and recommendation ranking.
- Repeated-failure academic exclusion (for example failing a module twice) cannot be represented by the current student profile, so no case tests it. It is recorded as a known limitation.

8. Next steps

- Record the academic reviewer and grounding judge, with committed hours, in the personnel file.

- Have the reviewer label the 60 blinded cases and draft concealed cases, then run adjudication.
- Repeat the runs with three passes per case on the pinned model, and measure the Gemini-assisted CurricuPath path.
- Ask the reviewer to settle whether a planned corequisite satisfies Rule 1.9, and record the outcome as a rule decision.
- Score grounding and the clarification ablation, then rebuild this report from the official outputs.

Appendix. Reproducing this report

All numbers in this document were read from the analysis files. To regenerate them:

py scripts/run_evaluation.py --arm rules_only --allow-draft
py scripts/run_evaluation.py --arm curricupath --allow-draft --runs 1 (with GEMINI_API_KEY blank)
py scripts/run_evaluation.py --arm handbook_rag --allow-draft --runs 1 --pace-seconds 3.5
py scripts/run_intent_evaluation.py
py scripts/score_evaluation.py --allow-draft --reference-sets developer briefing
py scripts/render_benchmark_figures.py
py scripts/build_evaluation_report.py

Input fingerprints (SHA-256, first 16 characters)

draft_curricupath.jsonl: 4665c9855595fc5c
draft_curricupath_no_clarification.jsonl: 7fa252ded5850982
draft_handbook_rag.jsonl: 2138e9d35e3020e3
draft_rules_only.jsonl: 0e723c46a10e9e57
draft_structured_rules_llm.jsonl: 5a3c3e4c4ae780bc
eligibility_cases.csv: 955bc130ded299b9
senate_briefing_labels.csv: ab33ec22d9f76467

Answer-key provenance

The briefing labels were extracted from *senate_rules_briefing.docx*, stored beside *senate_briefing_labels.csv* with its hash. The document describes every label as a draft and carries no reviewer name, signature or approval, so it is used for sensitivity analysis only.